

Total No. of Questions : 4]

SEAT No. :

PC396

[6359]-516

[Total No. of Pages : 2

S.E. (Electronics & Computer Engineering) (Insem)

ELECTRONIC CIRCUITS

(2019 Pattern) (Semester - III) (204202)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, & Q.3 or Q.4.
- 2) Figures to the right indicate full marks.
- 3) Draw neat diagram wherever necessary.
- 4) Assume suitable data, if necessary.

Q1) a) Explain construction and working of N channel Enhancement MOSFET. [5]

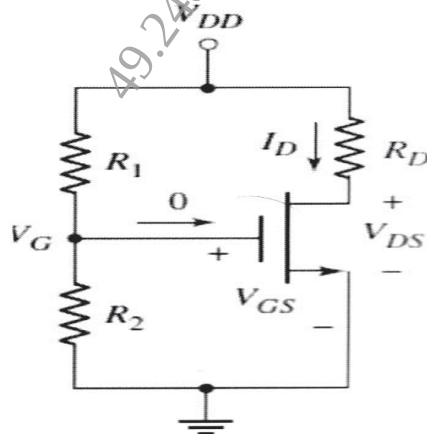
b) Explain the following parameters for MOSFET. [5]

i) Body Effect

ii) Subthreshold Conduction

c) Calculate the drain current and drain-to-source voltage of a common source circuit with an n-channel enhancement-mode MOSFET. [5]

Assume that $R_1 = 30k\Omega$, $R_2 = 20k\Omega$, $R_D = 20K\Omega$, $V_{DD} = 5V$, $V_{TN} = 1V$, and $K_n = 0.1 \text{ mA/V}^2$.



OR

P.T.O.

- Q2)** a) Draw and explain drain and transfer characteristics of N channel Enhancement MOSFET. [5]
- b) Explain the different breakdown effects in MOSFET. [5]
- c) Draw the circuit diagram of single stage CS amplifier and obtain it's AC equivalent circuit mentioning the rules to draw AC equivalent diagram. [5]

- Q3)** a) Draw MOSFET current source and explain in detail. [5]
- b) Compare the different negative feedback topologies for block diagram, input & output resistance. [5]
- c) Draw circuit diagram of RC phase shift oscillator and calculate the operating frequency if $R = 10k\Omega$ and $C = 5nF$. [5]

OR

- Q4)** a) Draw MOSFET current mirror circuit and explain in detail. [5]
- b) Enlist the advantages of negative feedback. [5]
- c) Draw circuit diagram of wein bridge oscillator and explain in detail. What is the frequency of oscillation? [5]

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